Research Proposal

Hierarchical Uncertainty Quantification for Automated Chest X-Ray Report Generation Using Vision-Language Transformers

1st Author: Abdul Samad

2nd Author: Zain Ali

Department of Software Engineering
Institute of Management Sciences
Peshawar, Pakistan

Course: Artificial Intelligence

Semester: 4th

Date: February 27, 2026

1. Introduction

Chest radiography is the most frequently performed diagnostic imaging procedure globally, with over 2 billion examinations conducted annually. Each chest X-ray requires interpretation by a trained radiologist who generates a comprehensive textual report describing anatomical observations, pathological findings, diagnostic impressions, and clinical recommendations. This labor-intensive process creates significant healthcare bottlenecks, particularly in developing regions facing critical radiologist shortages.

The shortage of radiologists is especially severe in low-resource countries. Pakistan maintains a radiologist-to-population ratio of approximately 1:50,000, compared to 1:10,000 in developed nations. This disparity results in diagnostic delays, increased workload related errors, and compromised patient outcomes. The COVID-19 pandemic further exacerbated these challenges, with radiology departments experiencing unprecedented imaging volumes while operating under constrained staffing conditions.

Recent advances in deep learning have demonstrated remarkable capabilities in medical image analysis. Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) achieve diagnostic accuracy comparable to human radiologists in binary classification tasks such as pneumonia detection, tuberculosis screening, and lung nodule identification. However, clinical radiology demands far more than simple yes/no predictions. Radiologists must produce structured, multi-sentence reports containing detailed anatomical descriptions, pathological findings with appropriate medical terminology, disease severity assessments, and follow-up recommendations—all while maintaining linguistic quality and clinical coherence.

Vision-Language Models (VLMs) have emerged as a promising solution for bridging computer vision and natural language generation. By combining image encoders with language decoders through attention mechanisms, VLMs can automatically generate descriptive text from images. Recent research has adapted these architectures for medical report generation, achieving promising results on benchmark datasets. However, existing systems suffer from a critical limitation: **the absence of uncertainty quantification**.

Current automated report generation models produce deterministic outputs without indicating confidence levels, making it impossible for clinicians to distinguish between high-confidence predictions and uncertain hypotheses. In high-stakes medical applications, this opacity poses significant risks. A model that confidently misdiagnoses a life-threatening condition is more dangerous than one expressing appropriate doubt and triggering human review. Furthermore, without uncertainty estimates, healthcare institutions cannot systematically identify cases requiring priority expert review.

Recent developments in Bayesian deep learning have demonstrated that neural networks can estimate prediction confidence through techniques such as Monte Carlo

Dropout, Deep Ensembles, and Evidential Deep Learning. However, these methods have primarily been applied to classification tasks, with limited exploration in generative models for medical text generation. The few existing works incorporating uncertainty in report generation provide only coarse, report-level confidence scores, failing to identify which specific sentences, phrases, or medical findings are uncertain.

This research addresses these limitations by developing a hierarchical uncertainty aware vision-language framework for chest X-ray report generation. Our approach integrates Vision Transformers for robust visual encoding, GPT-2 for fluent language generation, cross-modal attention for vision-language alignment, and Monte Carlo Dropout with Bayesian inference for principled uncertainty quantification at word, sentence, and section levels. Additionally, we incorporate attention visualization to provide explainable predictions, enabling clinicians to understand which image regions influenced specific textual findings.

The proposed system advances medical AI trustworthiness by enabling uncertainty guided clinical workflows where high-confidence predictions can be auto-validated while low-confidence cases are flagged for expert review. This hybrid human-AI approach optimizes radiologist productivity without compromising diagnostic safety, representing a critical step toward clinically deployable automated radiology systems for resource constrained healthcare settings.

2. Problem Statement

Despite significant advances in deep learning-based medical image analysis, existing automated chest X-ray report generation systems suffer from three critical limitations:

First, current models lack uncertainty quantification mechanisms. They generate deterministic textual outputs without providing confidence scores, making it impossible for clinicians to assess prediction reliability. This opacity prevents safe clinical deployment, as healthcare professionals cannot distinguish between confident diagnoses and uncertain hypotheses requiring manual verification.

Second, existing uncertainty-aware medical AI systems provide only coarse-grained confidence estimates (report-level or image-level), lacking the fine-grained word-level and sentence-level granularity required for actionable clinical decision-making. Radiologists need to know precisely which findings are uncertain, not merely that ”the overall report has 75% confidence.”

Third, most automated report generation models produce unstructured free-text paragraphs rather than adhering to standard clinical report format with distinct sections (Findings, Impression, Recommendations). This deviation from established radiological practice reduces report usability and clinical adoption.

Research Question: How can we develop an automated chest X-ray report generation system that produces clinically accurate, structured radiology reports while quantifying prediction uncertainty at hierarchical levels (word, sentence, section) to enable safe clinical deployment and effective human-AI collaboration in resource-constrained healthcare settings?

3. Literature Review

3.1 Medical Report Generation and Image Captioning

Automated medical report generation has evolved significantly over the past decade. Early template-based approaches retrieved pre-defined sentence templates based on image classification results, offering interpretability but lacking linguistic flexibility and variability.

Jing et al. (2018) introduced hierarchical LSTM models with co-attention mechanisms for chest X-ray report generation, achieving BLEU-4 scores of 0.217 on the IU X-Ray dataset. Their multi-level attention mechanism enabled the model to focus on different anatomical regions when generating distinct report sections. However, the system relied on CNN-based image encoders, which have since been superseded by Vision Transformers.

Li et al. (2019) proposed knowledge-driven approaches incorporating medical knowledge graphs (RadLex, UMLS) to improve clinical accuracy, reducing hallucinated findings by 23%. Despite improvements, their system generated unstructured paragraphs rather than section-based reports, limiting clinical utility.

Chen et al. (2020) developed memory-driven transformers storing frequently occurring disease patterns in external memory modules. This retrieval-augmented approach achieved state-of-the-art BLEU scores (0.353 on IU X-Ray). However, the memory retrieval process lacks uncertainty quantification, preventing the system from indicating when encountering novel or ambiguous cases.

Wang et al. (2022) combined Vision Transformers with meshed-memory transformers, achieving BLEU-4 of 0.392 on MIMIC-CXR. Their architecture processes images as patch sequences, enabling fine-grained attention over anatomical regions. Nevertheless, they provide no confidence estimates for generated text, limiting clinical trustworthiness.

3.2 Uncertainty Quantification in Deep Learning

Bayesian deep learning provides principled frameworks for uncertainty estimation in neural networks. **Gal and Ghahramani (2016)** demonstrated that dropout training approximates Bayesian inference. By performing multiple stochastic forward passes with dropout enabled during inference (Monte Carlo Dropout), they estimated predictive uncertainty through output variance. This technique has been successfully applied to medical image classification, achieving well-calibrated confidence scores.

Lakshminarayanan et al. (2017) proposed Deep Ensembles, training multiple networks with different initializations and averaging predictions. While effective for uncertainty estimation, this approach incurs significant computational overhead (N× model parameters), making it impractical for large vision-language models in real-time clinical settings.

Sensoy et al. (2018) introduced Evidential Deep Learning, placing Dirichlet distributions over class probabilities to model both aleatoric (data) and epistemic (model) uncertainty. However, this method is primarily designed for classification tasks with limited applicability to sequence generation problems.

In medical imaging applications, **Begoli et al. (2019)** demonstrated that uncertainty aware systems improve clinician trust and diagnostic accuracy. Their clinical study showed radiologists presented with AI predictions plus uncertainty scores made 18% fewer diagnostic errors compared to deterministic predictions alone, highlighting the critical importance of confidence estimates in medical AI deployment.

3.3 Uncertainty in Natural Language Generation

Uncertainty quantification for natural language generation remains an open research challenge. **Malinin and Gales (2021)** applied ensemble methods to neural machine translation, estimating uncertainty through output diversity across ensemble members. However, ensemble-based approaches are computationally prohibitive for real-time clinical applications requiring rapid report generation.

Xiao and Wang (2021) proposed sequence-level uncertainty estimation for abstractive summarization using Monte Carlo Dropout. They achieved moderate correlation between uncertainty scores and ROUGE metric degradation. However, their approach provides only sentence-level granularity, lacking word-level precision needed for identifying specific uncertain medical terms or findings.

Wang et al. (2024) represents the first work attempting uncertainty quantification specifically for medical report generation. Using variational autoencoders, they estimate

report-level confidence scores on MIMIC-CXR dataset. While promising, their coarse-grained uncertainty (one confidence value per entire report) lacks actionable clinical granularity. A radiologist needs precise information about which specific findings are uncertain, not an overall report confidence score.

3.4 Vision-Language Models and Cross-Modal Attention

Attention mechanisms enable models to focus on relevant input regions when generating output tokens. **Xu et al. (2015)** introduced spatial attention for image captioning, visualizing which image regions influence each generated word. This concept has been extended to medical imaging through cross-modal attention, aligning visual features with semantic medical concepts.

Recent large-scale vision-language models like CLIP (Radford et al., 2021) and BLIP (Li et al., 2022) demonstrate that pre-training on massive image-text pairs significantly improves downstream task performance. However, these general-domain models have not been systematically evaluated for uncertainty quantification in specialized medical applications requiring high reliability and safety standards.

3.5 Research Gap Identification

Despite substantial progress in automated medical report generation and uncertainty quantification, existing literature exhibits four critical gaps:

Gap 1: Coarse Uncertainty Granularity – Current uncertainty-aware systems (Wang et al., 2024) provide only report-level confidence scores, lacking fine-grained word-level and sentence-level estimates required for actionable clinical decision making. Radiologists need to know precisely which findings are uncertain, not overall report confidence.

Gap 2: Unstructured Report Generation – Most existing models (Chen et al., 2020; Jing et al., 2018) generate free-form paragraphs without adhering to standard radiology report structure (Technique, Findings, Impression, Recommendations). This deviation from clinical practice reduces report usability and adoption.

Gap 3: Attention Without Uncertainty Integration – Existing attention visualization methods (Wang et al., 2022; Li et al., 2019) show which image regions the model attends to but not how confident the model is about those regions. This prevents clinicians from assessing whether the model's visual focus is reliable or spurious.

Gap 4: Lack of Calibration Validation – Few studies rigorously validate whether predicted uncertainty correlates with actual model errors. Without calibration analysis, confidence scores may be unreliable, undermining clinical trust and safety.

Our work addresses these gaps by developing a hierarchical uncertainty quantification framework generating structured radiology reports with word, sentence, and section-level confidence scores, integrating uncertainty with attention visualization, and rigorously evaluating calibration quality against ground-truth diagnostic errors.

4. Research Objectives

The primary objectives of this research are:

1. **Architecture Development:** Design and implement a multimodal vision-language transformer architecture integrating Vision Transformer (ViT) for image encoding, GPT-2 for language generation, and cross-modal attention mechanisms for vision language alignment.

2. **Hierarchical Uncertainty Quantification:** Develop and evaluate Monte Carlo Dropout-based uncertainty estimation at three granularities:

    - *Word-level:* Confidence scores for individual medical terms and findings
    - *Sentence-level:* Uncertainty estimates for complete clinical statements
    - *Section-level:* Overall confidence for report sections (Findings, Impression, Recommendations)

3. **Structured Report Generation:** Generate radiology reports adhering to standard clinical format with distinct sections rather than unstructured paragraphs, matching real-world radiological practice.

4. **Uncertainty-Guided Explainability:** Implement attention visualization techniques combining visual attention heatmaps with uncertainty scores, enabling clinicians to understand both what the model observed and confidence levels for those observations.

5. **Comprehensive Evaluation:** Evaluate the system on standard benchmarks (IU X-Ray, MIMIC-CXR) using both NLP metrics (BLEU, METEOR, CIDEr) and clinical accuracy measures (precision, recall, F1-score for 14 common thoracic diseases).

6. **Uncertainty Calibration Analysis:** Assess whether predicted uncertainty scores correlate with actual model errors, ensuring high-uncertainty predictions genuinely indicate higher error likelihood.

5. Methodology

5.1 Dataset Selection

We will utilize two widely-used benchmark datasets for chest X-ray report generation:

1. IU X-Ray Dataset: Contains 7,470 chest X-ray images from Indiana University with corresponding radiology reports. Each report includes multiple sections (Findings, Impression) written by experienced radiologists. This dataset provides high-quality, structured reports suitable for training section-aware generation models.

2. MIMIC-CXR Dataset: Comprises 377,110 chest X-ray images from Beth Israel Deaconess Medical Center with free-text radiology reports. This large-scale dataset enables robust model training and generalization evaluation. Reports contain annotations for 14 common thoracic diseases including pneumonia, cardiomegaly, pneumothorax, and pleural effusion.

5.2 Data Preprocessing Image

Preprocessing:

- Resize all images to 224×224 pixels (ViT input requirement)
- Normalize pixel values to [0, 1] range
- Apply data augmentation: random horizontal flipping, rotation (±10°), brightness/contrast adjustment
- Convert DICOM format to PNG/JPG for efficient processing **Text Preprocessing:**
- Tokenize reports using Byte-Pair Encoding (BPE) with vocabulary size 50,257
- Insert section markers: <TECHNIQUE>, <FINDINGS>, <IMPRESSION>
- Add special tokens: <BOS> (beginning), <EOS> (end), <PAD> (padding)
- Remove PHI (Protected Health Information) and standardize medical terminology

- Handle class imbalance using weighted sampling for rare disease findings **Train-Test Split:**
- Training set: 70% (5,229 images for IU X-Ray)
- Validation set: 15% (1,120 images)
- Test set: 15% (1,121 images)
- Stratified splitting ensures balanced disease distribution across splits

5.3 System Architecture

Our proposed framework consists of four integrated components:

5.3.1 Component 1: Vision Transformer (ViT) Encoder

We employ a pre-trained Vision Transformer (ViT-Base/16) as our image encoder. Unlike CNNs processing images through hierarchical convolutions, ViTs partition images into fixed-size patches and process them as sequences using self-attention mechanisms.

Patch Embedding: Given a chest X-ray image $\mathbf{x} \in R^{224\times224\times1}$, we partition it into $N = 196$ non-overlapping patches of size 16 × 16. Each patch is linearly projected into a 768-dimensional embedding:

$$\mathbf{z}^0 = [\mathbf{x}_{cls}; \mathbf{x}_1^p\mathbf{E}; \mathbf{x}_2^p\mathbf{E}; \ldots; \mathbf{x}_N^p\mathbf{E}] + \mathbf{E}_{pos} \quad (1)$$

where $\mathbf{E} \in R^{256\times768}$ is the patch embedding matrix, $\mathbf{x}_{cls}$ is a learnable classification token, and $\mathbf{E}_{pos} \in R^{197\times768}$ represents position embeddings.

Transformer Encoding: Embedded patches are processed through 12 Transformer encoder layers using Multi-Head Self-Attention (MHSA) and feedforward networks. The output visual features $\mathbf{V} = \mathbf{z}_{12} \in R^{196\times768}$ encode spatial information about anatomical structures and pathological findings.

5.3.2 Component 2: GPT-2 Language Decoder

We fine-tune GPT-2-Medium (355M parameters) as our language decoder for autoregressive text generation. GPT-2 employs a unidirectional Transformer architecture optimized for fluent, coherent text production.

Autoregressive Generation: Given visual features $\mathbf{V}$ and previously generated tokens $\mathbf{y}_{<t}$, GPT-2 predicts the next token distribution:

$$p(y_t|\mathbf{y}_{<t},\mathbf{V}) = \text{softmax}(\mathbf{W}_o\mathbf{h}_t) \quad (2)$$

where $\mathbf{h}_t$ is the hidden state at timestep t and $\mathbf{W}_o$ is the output projection matrix.

5.3.3 Component 3: Cross-Modal Attention Bridge

To integrate visual information into language generation, we insert cross-modal attention layers between GPT-2's self-attention and feedforward layers. This allows the language model to attend to relevant image regions when generating specific medical terms.

Cross-Attention Mechanism: At each decoding timestep t, the language model hidden state (query) attends to visual features (keys and values):

$$\text{Attention}(\mathbf{Q},\mathbf{K},\mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right)\mathbf{V} \quad (3)$$

Multi-head cross-attention applies this mechanism across 8 parallel attention heads, enabling simultaneous focus on different anatomical regions.

5.3.4 Component 4: Monte Carlo Dropout for Uncertainty Quantification

We employ Monte Carlo Dropout to estimate epistemic uncertainty at multiple granularities. During inference, instead of a single forward pass, we perform $T = 20$ stochastic forward passes with dropout enabled (dropout rate $p = 0.2$).

Word-Level Uncertainty: For each word w_t, we compute:

Predictive Mean:

$$\bar{p}(y_t) = \frac{1}{T}\sum_{i=1}^{T} p_i(y_t|\mathbf{y}_{<t}, \mathbf{V}) \quad (4)$$

Predictive Entropy (Uncertainty):

$$H(y_t) = -\sum_{v\in V} \bar{p}(v)\log \bar{p}(v) \quad (5)$$

Confidence Score:

$$\text{Confidence}(w_t) = 1 - \frac{\mathcal{H}(y_t)}{\log|\mathcal{V}|} \quad (6)$$

Sentence-Level Uncertainty: For sentence $S = (w_1,\ldots,w_n)$:

$$\text{Confidence}(S) = \left(\prod_{t=1}^{n} \text{Confidence}(w_t)\right)^{1/n} \quad (7)$$

Section-Level Uncertainty: For section $S = \{S_1,\ldots,S_m\}$:

$$\text{Confidence}(\mathcal{S}) = \frac{1}{m}\sum_{j=1}^{m}\text{Confidence}(S_j) \quad (8)$$

5.4 Training Procedure

Pre-training:

- Vision Transformer: Pre-trained on ImageNet-21k, fine-tuned on ChestX-ray14
- GPT-2: Pre-trained on web text, fine-tuned on PubMed medical literature

Fine-tuning:

- Loss function: Cross-entropy over generated tokens
- Optimizer: AdamW with learning rate 3×10^{-5}
- Batch size: 16 image-report pairs
- Training epochs: 50 with early stopping (patience = 5)
- Hardware: NVIDIA A100 GPU (40GB VRAM)
- Estimated training time: 72 hours

5.5 Evaluation Metrics

Natural Language Generation Metrics:

- BLEU-1/2/3/4: N-gram overlap with reference reports
- METEOR: Accounts for synonyms and paraphrasing
- ROUGE-L: Longest common subsequence similarity • CIDEr: Consensus-based metric weighted by term frequency **Clinical Accuracy Metrics:**
- Micro/Macro-averaged Precision, Recall, F1-score for 14 thoracic diseases
- Area Under ROC Curve (AUROC) per disease
- Confusion matrices for multi-label classification **Uncertainty Quality Metrics:**

- Expected Calibration Error (ECE): Measures calibration quality
- Negative Log-Likelihood (NLL): Probabilistic forecast quality
- Brier Score: Accuracy of probabilistic predictions
- Selective Prediction Analysis: Accuracy vs. coverage trade-off

5.6 Baseline Comparisons

We will compare our approach against five established methods:

1. Show, Attend and Tell (2015): CNN + LSTM with attention
2. R2Gen (2020): Relational memory-based generation
3. CMN (2021): Cross-modal memory network
4. Meshed-Memory Transformer (2020): Memory-augmented transformer
5. Wang et al. (2024): Current SOTA with report-level uncertainty

5.7 System Architecture Diagram

6. Expected Outcomes

Based on preliminary experiments and literature analysis, we anticipate the following outcomes:

6.1 Quantitative Performance

Table 1: Expected Performance Comparison on IU X-Ray Test Set

Model	BLEU-4	METEOR	CIDEr	F1-Score
Show, Attend, Tell	0.217	0.185	0.421	0.532
R2Gen	0.304	0.248	0.583	0.614
CMN	0.328	0.267	0.612	0.641
Wang et al. (2024)	0.353	0.284	0.638	0.667
Our Model	**0.365-0.375**	**0.290-0.298**	**0.650-0.665**	**0.675-0.688**

We expect our model to achieve 3-5% improvement over current state-of-the-art in BLEU and CIDEr scores while providing hierarchical uncertainty quantification absent in existing approaches.

6.2 Uncertainty Calibration

- **Expected Calibration Error (ECE):** ¡ 0.10 (well-calibrated predictions)
- **Selective Prediction:** Rejecting bottom 20% confident predictions should increase accuracy by 8-12%
- **Uncertainty-Error Correlation:** Spearman correlation coefficient $\rho > 0.70$ between predicted uncertainty and actual model errors

6.3 Qualitative Outcomes

- **Structured Reports:** Over 95% of the reports generated will correctly separate Findings and Impression sections matching clinical format
- **Uncertainty Visualization:** Attention heatmaps will clearly highlight uncertain anatomical regions requiring manual review
- **Clinical Utility:** Low-confidence predictions flagged for radiologist review will demonstrate higher error rates, validating uncertainty-guided workflow effectiveness

7. Significance of the Study

This research makes several significant contributions to medical AI and healthcare analytics:

7.1 Scientific Contributions

- **Novel Framework:** First hierarchical uncertainty quantification system for medical report generation operating at word, sentence, and section granularities
- **Methodological Innovation:** Integration of uncertainty estimates with attention visualization for explainable, confidence-aware predictions
- **Calibration Analysis:** Rigorous validation demonstrating predicted uncertainty correlates with actual diagnostic errors

7.2 Clinical Impact

- **Patient Safety:** Enables safer AI deployment by automatically flagging uncertain predictions requiring human review, reducing misdiagnosis risk

- **Workflow Optimization:** Radiologists can prioritize uncertain cases while auto validating high-confidence predictions, improving efficiency
- **Healthcare Accessibility:** Addresses radiologist shortage in developing regions like Pakistan, improving diagnostic access for underserved populations

7.3 Educational Impact • Provides hands-on experience with cutting-edge vision-language models, Bayesian deep learning, and medical AI systems • Develops critical research competencies: literature synthesis, experimental design, statistical analysis, academic writing

- Prepares students for advanced graduate research or industry careers in AI/healthcare technology sectors

8. Conclusion

Automated chest X-ray report generation represents a promising application of artificial intelligence to address critical healthcare challenges. However, existing systems lack uncertainty quantification mechanisms essential for safe clinical deployment. This research proposes a novel hierarchical uncertainty-aware vision-language framework combining Vision Transformers, GPT-2 language models, cross-modal attention, and Monte Carlo Dropout-based Bayesian inference.

By generating structured radiology reports with word, sentence, and section-level confidence scores, our approach enables uncertainty-guided clinical workflows where high confidence predictions are auto-validated while uncertain cases are flagged for expert review. This hybrid human-AI collaboration optimizes radiologist productivity without compromising diagnostic safety—a critical requirement for real-world medical AI deployment.

The proposed system will be rigorously evaluated on standard benchmarks (IU XRay, MIMIC-CXR) using both NLP metrics and clinical accuracy measures. We expect to achieve competitive or superior performance compared to existing methods while providing actionable uncertainty estimates absent in current state-of-the-art systems. This work advances trustworthy medical AI by demonstrating that uncertainty quantification can transform automated diagnostic systems from opaque black boxes into transparent, confidence-aware clinical decision support tools suitable for resource-constrained healthcare settings.

9. References

Base Paper

Title: Uncertainty Quantification for Automated Radiology Report Generation
Authors: Wang, H., Zhang, L., Chen, Y., and Liu, X.
Year / Publication: 2024, IEEE Transactions on Medical Imaging

Key Contribution: Wang et al. (2024) proposed the first uncertainty-aware medical report generation system using variational autoencoders on MIMIC-CXR dataset. Their model provides report-level confidence scores achieving 0.353 BLEU-4 with 78% average confidence correlation. However, their coarse-grained uncertainty (one score per report) lacks clinical actionability. Our work extends this by introducing hierarchical word, sentence, and section-level uncertainty quantification integrated with attention visualization for fine-grained confidence assessment.

Additional Key References

1. Jing, B., Xie, P., and Xing, E. (2018). "On the Automatic Generation of Medical Imaging Reports," *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (ACL)*, pp. 2577-2586.

2. Li, Y., Liang, X., Hu, Z., and Xing, E. (2019). "Hybrid Retrieval-Generation Reinforced Agent for Medical Image Report Generation," *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 32.

3. Chen, Z., Song, Y., Chang, T., and Wan, X. (2020). "Generating Radiology Reportsvia Memory-Driven Transformer," *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 1439-1449.

4. Gal, Y., and Ghahramani, Z. (2016). "Dropout as a Bayesian Approximation: Representing Model Uncertainty in Deep Learning," *Proceedings of the 33rd International Conference on Machine Learning (ICML)*, pp. 1050-1059.

5. Lakshminarayanan, B., Pritzel, A., and Blundell, C. (2017). "Simple and ScalablePredictive Uncertainty Estimation using Deep Ensembles," *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30.

6. Begoli, E., Bhattacharya, T., and Kusnezov, D. (2019). "The Need for UncertaintyQuantification in Machine-Assisted Medical Decision Making," *Nature Machine Intelligence*, vol. 1, no. 1, pp. 20-23.

7. Dosovitskiy, A., et al. (2021). ”An Image is Worth 16×16 Words: Transformers for Image Recognition at Scale,” *International Conference on Learning Representations (ICLR)*.

8. Radford, A., Wu, J., Child, R., et al. (2019). ”Language Models are UnsupervisedMultitask Learners,” OpenAI Technical Report.

9. Xu, K., Ba, J., Kiros, R., et al. (2015). ”Show, Attend and Tell: Neural Image CaptionGeneration with Visual Attention,” *Proceedings of the 32nd International Conference on Machine Learning (ICML)*, pp. 2048-2057.

10. Rajpurkar, P., Irvin, J., Ball, R. L., et al. (2017). ”CheXNet: Radiologist-Level Pneumonia Detection on Chest X-Rays with Deep Neural Networks,” *arXiv preprint arXiv:1711.05225*.

11. Malinin, A., and Gales, M. (2021). ”Uncertainty Estimation in Autoregressive Structured Prediction,” *International Conference on Learning Representations (ICLR)*.

12. Sensoy, M., Kaplan, L., and Kandemir, M. (2018). ”Evidential Deep Learning to Quantify Classification Uncertainty,” *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 31.